

JUDGING (for us)

Final Deliverables + judging criteria

1. **GitHub repository**
2. **2-minute business video**
3. **2-minute technical demo video**

1. **Originality** of the idea,
2. **Feasibility** of the solution
3. **Potential impact**
4. **Technical execution**
5. **Clarity** of the presentation
6. Connection with the **Hong Kong ecosystem.**

For the Hong Kong ecosystem part, this does **not** mean that your product has to be built only for Hong Kong. A global product can be a very strong project if it is also relevant and applicable to Hong Kong. What matters is that you can show why Hong Kong could be a meaningful market, launchpad, testbed, or gateway to Asia for your solution.

Neurotechnology ..

EuroTech x HKTE Hackathon * June 7

JvF * AgT * MmK * DtP

Background: Today

Consumer medical devices...

when we think of consumer medtech
we think of wearable devices constantly
monitoring and diagnosing you

CONSUMER

RESEARCH/CLINICAL

smart watches

the stupid ring

Consumer med devices make it so anyone has access to their...

HR, breathing, sleep quality, blood oxygen,
menstrual cycles, walking stability, ambient noise

Why not features from their brain?

Background: Today

Spectrum of Neurotechnology: biggest players

CONSUMER

RESEARCH/CLINICAL



HITACHI



muse

...

mindmaze
THERAPEUTICS



g. tec

invasive recording



FATIGUE MONITORING

Where are we going?

- Meta's 2019 acquisition of ctrl-labs for EMG wristbands

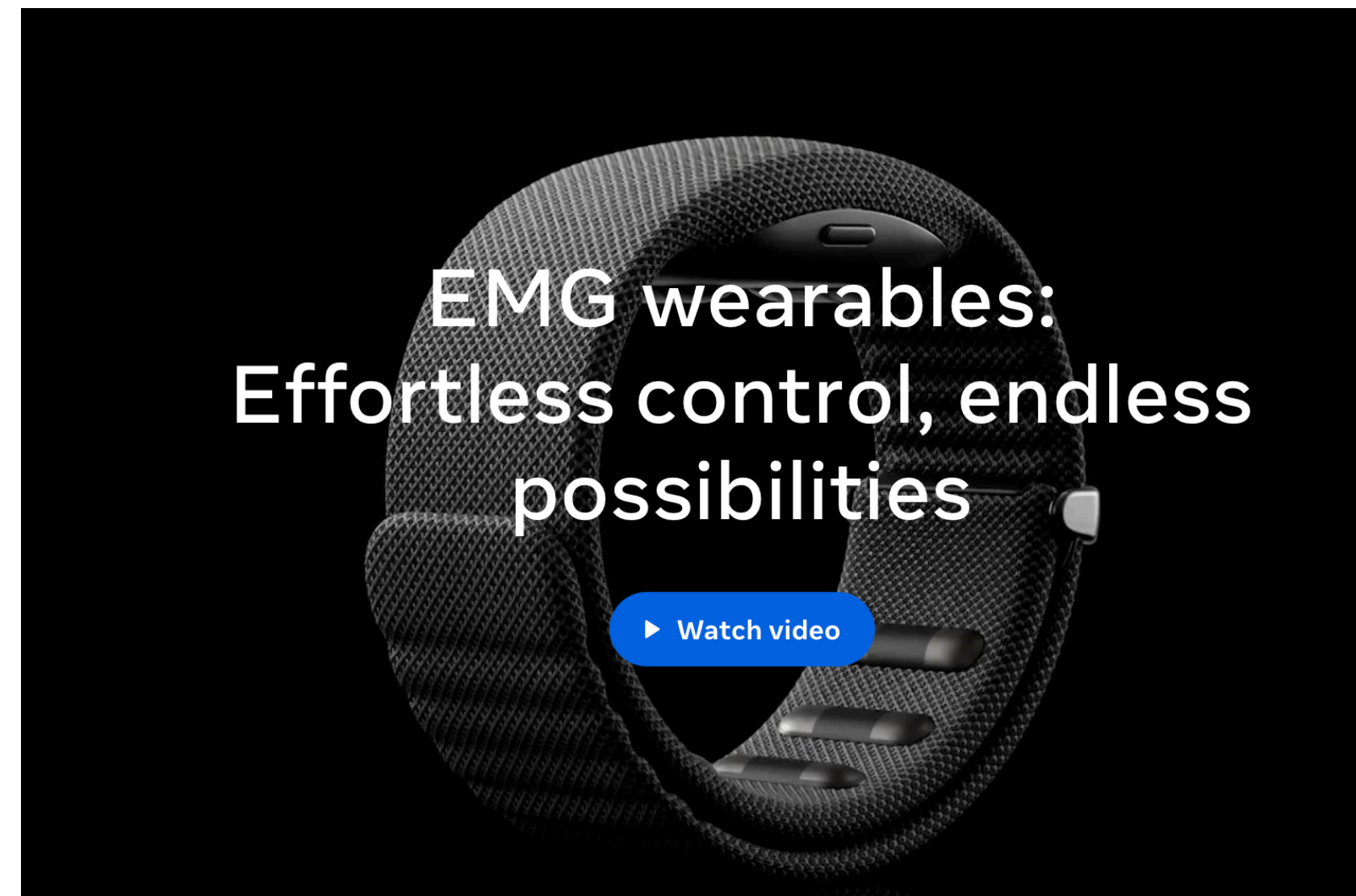
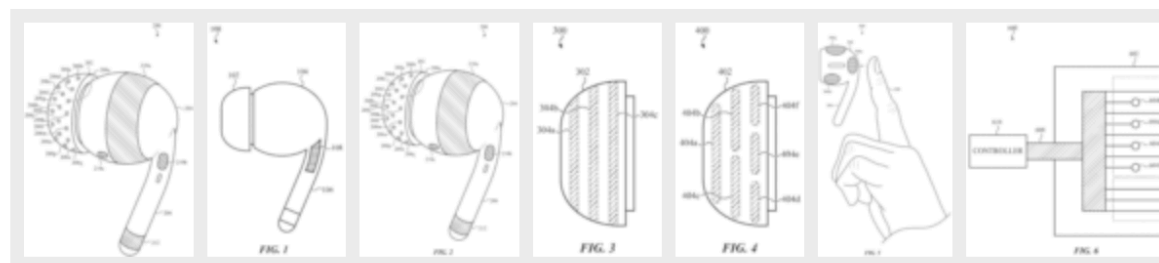
- Apple's 2023 In-ear EEG patent

Biosignal Sensing Device Using Dynamic Selection of Electrodes

Abstract

A wearable electronic device includes a housing, and an electrode carrier attached to the housing and having a nonplanar surface. The wearable electronic device includes a set of electrodes, including electrodes positioned at different locations on the nonplanar surface. The wearable electronic device includes a sensor circuit and a switching circuit. The switching circuit is operable to electrically connect a number of different subsets of one or more electrodes in the set of electrodes to the sensor circuit.

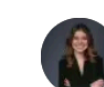
Images (15)



TECH

Elon Musk shows off updates to his brain chips and says he's going to install one in himself when they are ready

PUBLISHED THU, DEC 1 2022•8:09 AM EST | UPDATED THU, DEC 1 2022•10:36 AM EST



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- Musk's readiness in 2022 to get Neuralink implant

**We've been on the verge for years..
Why are there no consumer brain
computer interfaces?**

**No data.
No algorithms.
No application.**

We have all three.

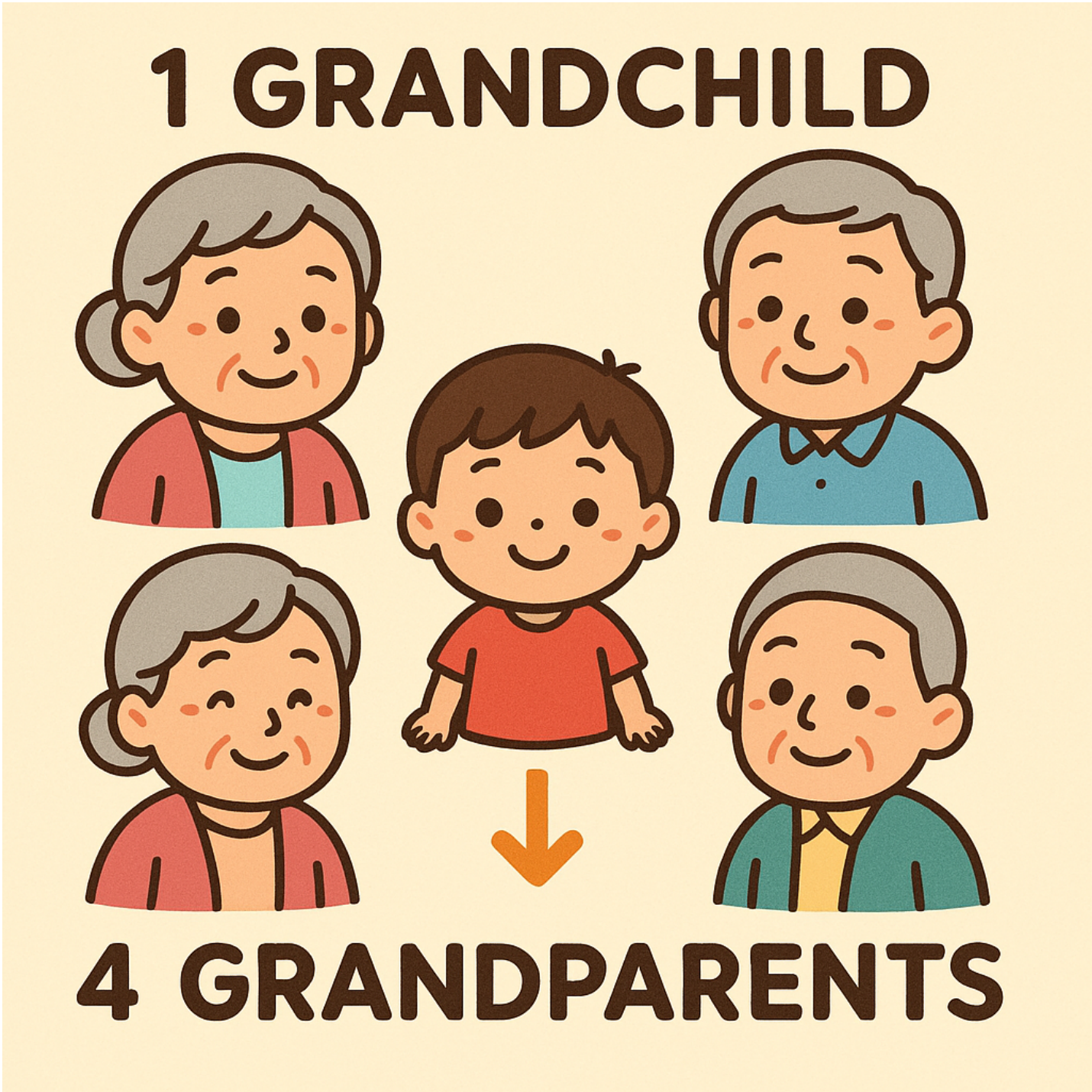
**Built around our novel AI for
generalizable non-invasive EEG
classification**

This is only possible now bc

1. There exists a market consumer / relatively cheaper EEG headset companies
2. EEG embedding models are fairly new so data is easier to store and use
3. We found a market

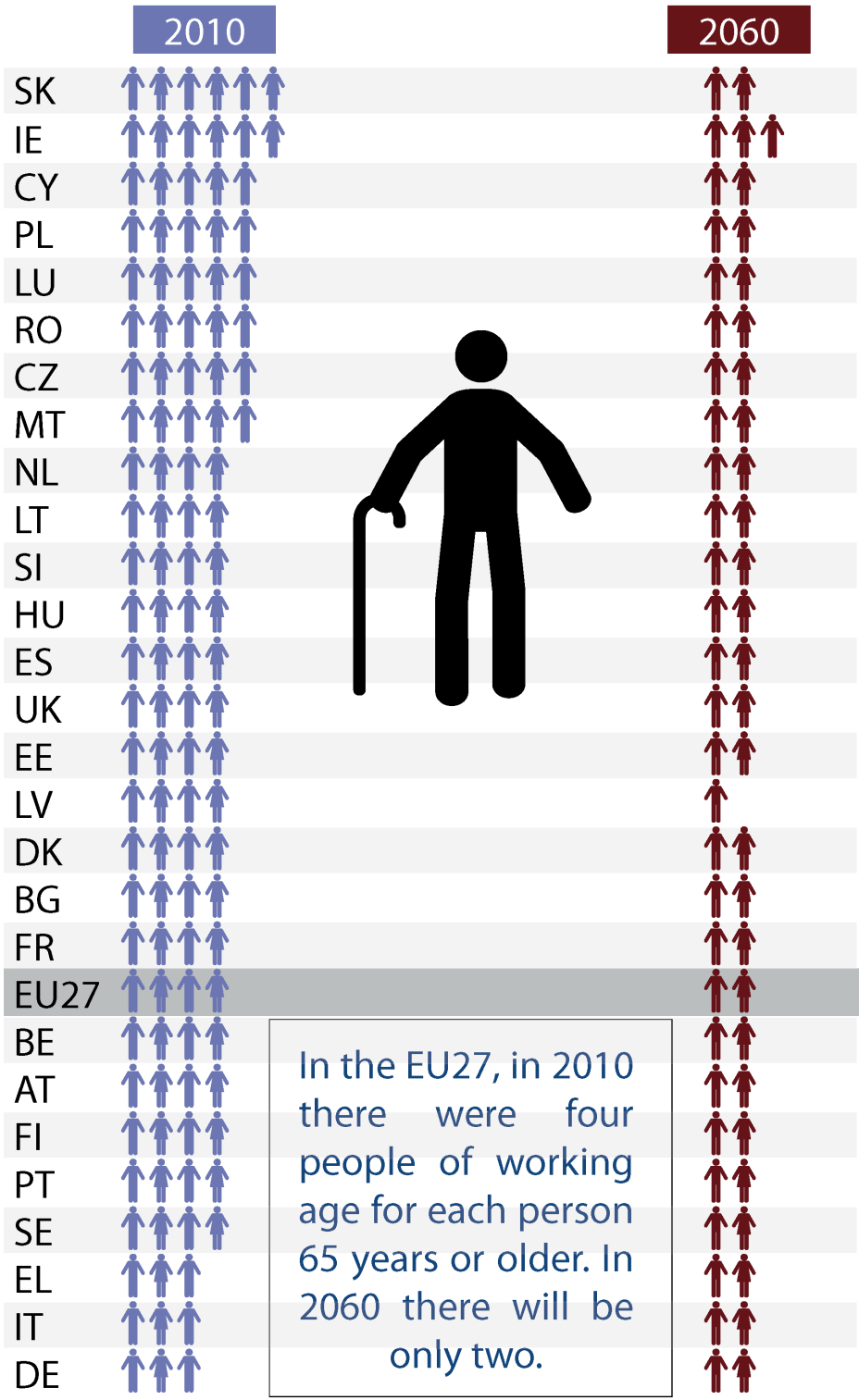
The Aging-Population problem

Europe, Asia, & the world



Old-age dependency ratio
(65+/(15-64))

Number of people of working age for each person 65 years or older



What does a caretaker do?

Clinical vs non-clinical

- *Find out what they do and show explicit parallels that EEG can assist in*

The app

Hardware agnostic self learning patient+caretaker app

description. pictures. features. idfk

The app

examples of things we can help

Business plan

1. Users

1. help people yay!!!

2. Data

1. **New labels** for rare events
2. Variety environments of **continuously streamed electrophysiological** data
3. integration with other medical recording (HR, breathing rate, movement, etc)

if we own the embedding model that means all
new classifiers will have to call our model...
something greater is at play here

Why Hong Kong?

1. The aging-population problem is big there (statistically + culturally *could be good to find some propaganda anthology stuff saying why this is such a problem there (they care abt their elders a lot)*)
2. China+HK already have a neurotech ecosystem
3. Since they already have the ecosystem, would benefit from more labeled EEG data opening doors for more advancements in EEG

I think the point has to be something like
(1) this will address a real problem in asia +
(2) this is the easiest place in the world to implement this

China neurotech ecosystem

Expansion

What can we do when we have more data?

- Neurological affects/monitoring with drug intake (from labels)
 - Finding deviations from normal or expected behavior -> quantifiable effect of medication
- Translation of other monitoring to brain data
 - New dimension of things your watch can see
- better understanding of brain and blah blah
-

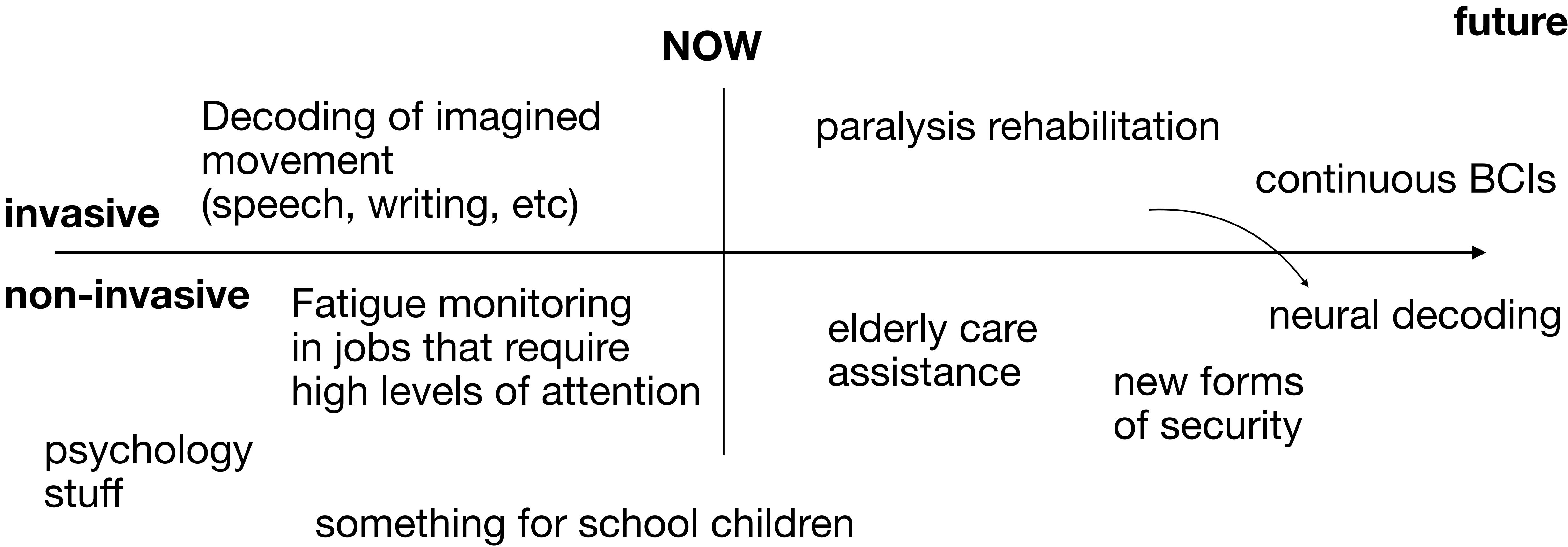
Apple Health W

eeg is more effective than
fingerprint/face ID for
passkey ... help prevent
fraudulent purchases against
elderly... dont know how to
incorporate this

Extension

don't love this i just want to say other ideas that will come from having more data idk

Where does this go next? Examples and ideas



concerns/expected questions

1. hardware? could be that they are our customers?
2. data model storage
- 3.